

REPORT

APPLICATION OVERVIEW

There are wide range of web applications to develop as web application is concerned, but our focus on this project, we are interested in developing a hotel management system. The system is created or developed to fulfill the following operations. The system allows the administrator to perform CRUD operations such as creating, editing, displaying and deleting data from the system. It is also allowing the user to view or check room details such as room number, room type, room price and status. Not only that, it is allowing the user and the administrator to identify the room's status either occupied or vacant once the room is taken.

The primary target user of this application is an administrator such as hotel manager, property agent or the receptionist who are responsible for the following duties, adding new rooms, viewing available rooms, updating room details such as room number, room type and room status. Another role is to delete room records that is not exit at that particular moment in time. This system is not developed for general users or customers but it is strictly designed for the administrative control and management.

DESIGN DECISION

We have decide to develop this system with the following technologys

- Frontend : The frontend of the system is created with html and css.
- Backend : The backend of the system is created with php.
- Database : The database that has been used in this system is MySQL

We have used this kind of technology because it is lightweight, easy to implement and ideal for beginners .php is well integrates well with MySQL database, making its easy to handle database operations .Html and css were used to design simple user friendly interface . These technologies are widely supported and can ideal on local servers such us XAMPP, making the development and testing process run smoothly

The many table in the database is apartment , which have the following fields such as:

- Id : which act as primary key field in the apartment table.
- roomNumber: which has been set unique field in the table.
- roomType: which express the condition of the room .
- price: amount to be paid .
- status : condition of the room

The CRUD operations were implemented as follows;

- create : the form allows admin to enter room details such as roomnumber , type,status and price which are directed to database using SQL INSERT query.

- Read :The room details accessed through SQL SELECT query and data is displayed in a table format on web page .
- Update: Admin can edit existing room details through edit button which sends updated data to the server through an SQL UPDATE query.
- Delete :room details can be erased from the database through SQL DELETE query initiated by delete button .

These operations are handled through separate php files , to maintain code readability and clarity .

Authentication was added to prevent unauthorized access to the system .This means that only the one who has access(Admin) can access the system .

- A login form: corrects the user and passwords
- Credentials : these credentials are validated against stored data in the database .
- If the credentials are correct a session is created using php sessions .
- Unauthorized are not given access to a session .

. Challenges & Solutions

There are a lot of things we have encountered during system development period some of them are database connection failure , php connection testing failure .

Solutions

- In terms of database connection failure ,we verified correct database name and table name ensuring that they match .
- In terms of php connection testing failure we ensured that both Apache and MySQL are all on

After correcting above challenges the system was successfully running.

Testing & Reflection

We carried out manual testing to ensure that everything is working smoothly .the system was tested in following areas:

- Adding room : we tried to add new room details to the table and it appeared
- Editing : room details were able to be altered .
- Delete : room details were able to be removed
- Connection : we tried to check if data is sent to the database .

All above testing was successfully done .

Had it been that we had enough time the system could have been able to allow customers to book rooms and other services such as conference halls ,dinning rooms and swimming pool.

During the system development period we have rent a number of things some of them are:

- Database connection : we have learnt how to connect php with database(MySQL) .
- SQL structure : we have learnt query languages such as INSERT ,DELETE and SELECT and how to use them .
- Database creation : we can now create database table using SQL query language and graphical driven language.